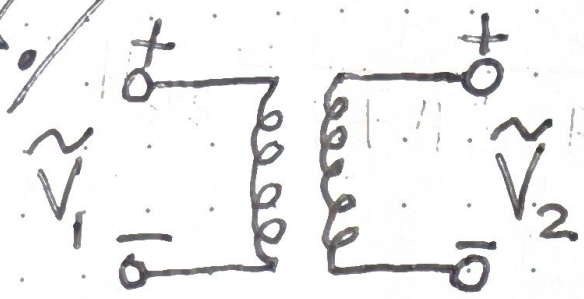
251119 Magnetically Coupled Circuits

ex./



$\tilde{V}_1 = 120V \quad \tilde{I}_1 = 1A$

$N_1 = 100 \quad N_2 = 10$

$\Rightarrow n = N_2/N_1 = 10/100 = 0.1$

$$\frac{\tilde{V}_2}{\tilde{V}_1} = n \Rightarrow 120(0.1) = \boxed{\tilde{V}_2 = 12V}$$

$$\frac{\tilde{I}_2}{\tilde{I}_1} = \frac{1}{n} \Rightarrow 1/(0.1) = \boxed{\tilde{I}_2 = 10A}$$

Check that Power is conserved

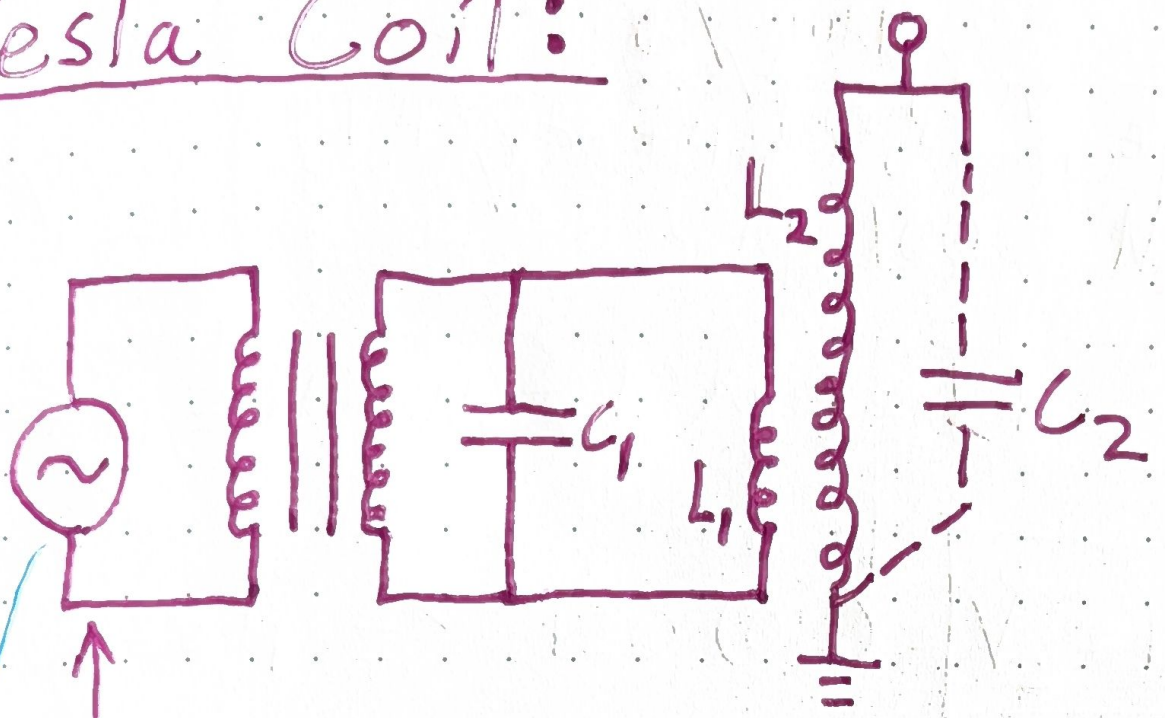
$P = IV$

$\Rightarrow P_1 = (120)(1) = 120W$

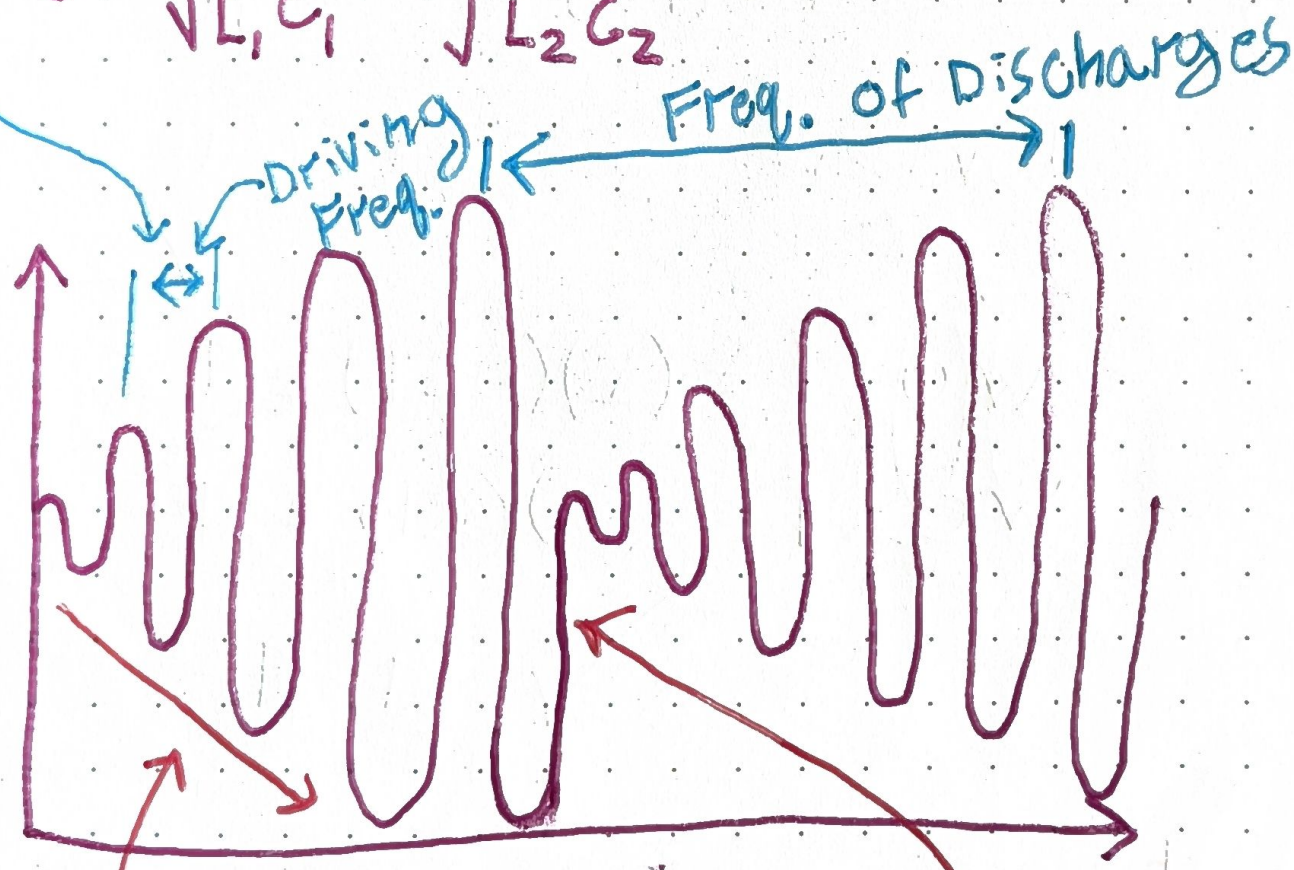
$\Rightarrow P_2 = (12)(10) = 120W$

$\therefore P_1 = P_2$ ✓

Tesla Coil:



$$\omega_0 = \frac{1}{\sqrt{L_1 C_1}} = \frac{1}{\sqrt{L_2 C_2}}$$



(Qualitative)

Increasing amplitude due to driving at resonence

Discharge (spark)